

Nicolas Serrano García

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Education

University of Alicante

BSc in Computer Engineering

Alicante, Spain

Sept. 2024 – Present

Honors Matrícula de Honor (highest distinction) in Computer Architecture and Programming & Data Structures.

ARA Track Selected for the English-taught track, limited to the top 25 students by admission score.

Average 8.0/10

University of Sydney

Study Abroad Program

Sydney, Australia

Feb. 2027 – Jun. 2027

Projects

ninttp

Personal Project

C++23 HTTP Library / C++, CMake, POSIX Sockets, Winsock, HTTP/1.1

- Developed a C++23 header-only HTTP client/server library with a C++-native API for desktop, embedded and constrained environments.
- Built a portable networking layer supporting POSIX sockets and Winsock, RAII ownership, IPv4/IPv6 and non-blocking I/O, designed to extend to RTOS, ESP32 and Nintendo 3DS.
- Implemented incremental HTTP/1.0 and HTTP/1.1 parsers with RFC-oriented validation and compile-time configurable limits.
- Added virtual-host, path and method routing, persistent connections, HTTP pipelining and graceful shutdown.
- Set up reusable CMake packaging and cross-platform CI across Windows/MSVC and Linux/GCC/Clang.

nincpu

Ongoing Personal Project

Digital CPU Design / Logisim Evolution, Computer Architecture

- Designing a simple CPU architecture with separate ALU, memory, control and integration circuits.
- Built arithmetic components including a full adder and 16-bit adder/subtractor.
- Developed register and memory subsystems using reusable 16-bit components.

DeskUp

Personal Project

Desktop Application / C++, Qt/SDL, X11, Win32, AppImage, Inno Setup

- Built a cross-platform desktop application to recover recently used workspace tabs.
- Integrated X11 and Win32 APIs and developed CI/CD workflows.
- Packaged releases for Linux and Windows using AppImage and Inno Setup.

Additional Learning

• *C++ Concurrency in Action* – Anthony Williams

Code lab

• *C++ Memory Management* – Patrice Roy

Code lab

Skills & Interests

Technical: C++, CMake, Google Test, Git, POSIX sockets, x86, MIPS, Qt, SDL, Doxygen, AppImage, Inno Setup

Languages: Spanish (native), English (C1 Cambridge), Valencian (C1)

Interests: Drums, camping